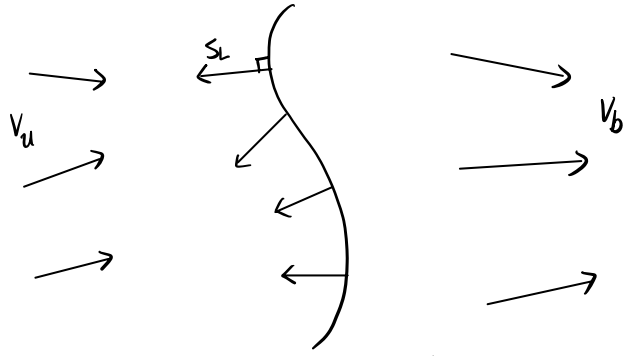
A field equation describing the flame position.

The kinematic relation $\boxed{\frac{dr_f}{dt}} = V_u + S_{L,u}$ → burning velocity

↓ flow velocity

↙ displacement velocity



unburnt mixture ($G < G_0$)

burnt gas ($G > G_0$)

Suppose that $\vec{n}$ is the normal vector to the flame surface

$$\Rightarrow \quad \frac{d\vec{x}_f}{dt} = \vec{v} + S_L \vec{n}$$

where $\vec{x}_f$ is the position vector of flame

$\frac{d\vec{x}_f}{dt}$ is called the flame propagation velocity

A common model to describe the propagation of interface is level set function $G(\vec{x}, t)$ as scalar field such that

$G(\vec{x}, t) = G_0$ represents the flame surface

$$\Rightarrow \quad \vec{n} = -\frac{\nabla G}{|\nabla G|}$$

where $\vec{n}$ is defined to point towards the unburnt mixture.

Differentiating $G(\vec{x}, t) = G_0$ with respect to t at $G = G_0$ gives

$$\frac{\partial G}{\partial t} + \nabla G \cdot \frac{\partial \vec{x}}{\partial t}\Big|_{G=G_0} = 0$$

$$\frac{\partial \vec{x}}{\partial t}\Big|_{G=G_0} = \frac{d\vec{x}_f}{dt} = \vec{V} + S_L \vec{n}$$

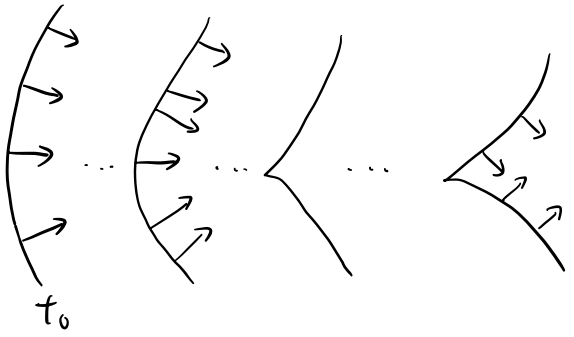
$$\frac{\partial G}{\partial t} + \vec{V} \cdot \nabla G = -S_L \vec{n} \cdot \nabla G$$

$$\vec{n} = -\frac{\nabla G}{|\nabla G|}$$

$$\boxed{\frac{\partial G}{\partial t} + \vec{V} \cdot \nabla G = S_L |\nabla G|}$$

According to Huygen's principle

As long as the interface is undulated in a quiescent flow, it will form cusps and eventually become flatter with time, whether it is smooth or not.

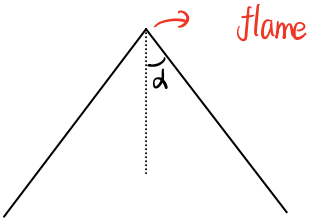


Special Case: Constant velocity, premixed, two-dimension

$\Rightarrow$ Axisymmetric Bunsen burner

$$\Rightarrow \quad u\frac{\partial G}{\partial x} = S_L \sqrt{\left(\frac{\partial G}{\partial x}\right)^2 + \left(\frac{\partial G}{\partial y}\right)^2}$$

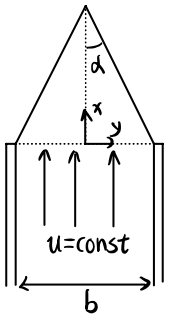
$\Rightarrow$



need a modification

$\Rightarrow$ flame stretch

Suppose that $G = x + F(y)$



$$\Rightarrow \quad u = S_L \sqrt{1 + \left(\frac{\partial F}{\partial y}\right)^2}$$

$$\Rightarrow \quad \left(\frac{\partial F}{\partial y}\right)^2 = \left(\frac{u}{S_L}\right)^2 - 1$$

$$\Rightarrow \quad F = \sqrt{\frac{u^2 - S_L^2}{S_L^2}}\,|y| + \text{const}$$

$$\Rightarrow \quad G = \sqrt{\frac{u^2 - S_L^2}{S_L^2}}\left(|y| - \frac{b}{2}\right) + x$$

Flame stretch

two contribution { Flame curvature
Flow divergence or strain

$$S_L = S_L^0 - S_L^0 \mathcal{L} K + \mathcal{L}\, n \cdot \nabla v \cdot n$$

where S_L^0 is the burning velocity for an unstretched flame

$\mathcal{L}$ is the Markstein length

$$\Rightarrow \quad \frac{S_L}{S_L^0} = 1 - \mathcal{L} K + \mathcal{L}\, \vec{n} \cdot \nabla \vec{v} \cdot \vec{n} \qquad \nabla \vec{v} \cdot \vec{n} = \frac{\partial v_i}{\partial x_i}$$

↓ neglect the flow divergence

$$\Rightarrow \quad \frac{S_L}{S_L^0} = 1 - \mathcal{L} K$$

→ flame thickness

$$\Rightarrow \quad \frac{S_L}{S_L^0} = 1 - \frac{L_F}{n} K$$

$$\Rightarrow \quad \frac{d(S_L/S_L^0)}{d(L_F K)} = -\frac{1}{n}$$

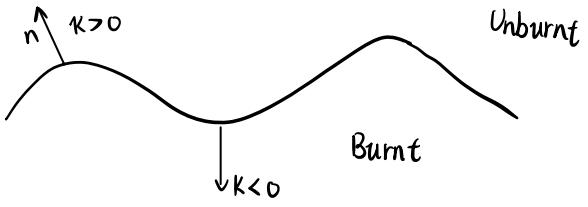
↘ Markstein number

$$K = \nabla \cdot n = -\nabla \cdot \left(\frac{\nabla G}{|\nabla G|}\right) = -\frac{\nabla^2 G + n \cdot \nabla (n \cdot \nabla G)}{|\nabla G|}$$

↘ flame curvature

Markstein length $\mathcal{L}$ is of the same order magnitude and proportional to laminar flame thickness l_F

Flame instabilities: Thermo-diffusive instability



$$S_L = S_L^0 - S_L^0 L \kappa - L S$$

$$S = -n_i \frac{\partial u_i}{\partial x_j} n_j$$ effect of stretch

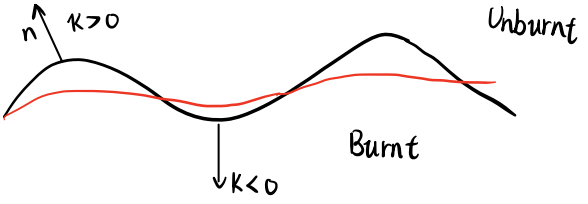
$$\kappa = \frac{\partial n_i}{\partial x_i}$$

effect of curvature

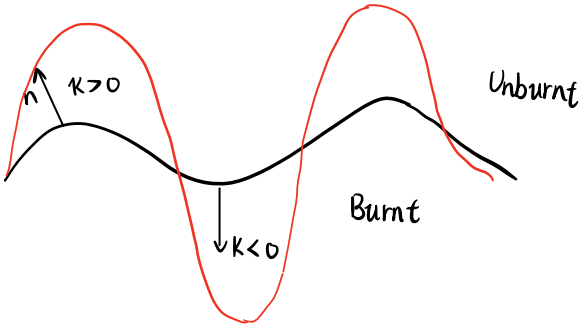
Only consider the curvature

$$S_L = S_L^0 - S_L^0 L \kappa$$

If $L > 0$, $\begin{cases} \kappa > 0 \Rightarrow S_L \downarrow \Rightarrow \text{move to burnt region} \\ \kappa < 0 \Rightarrow S_L \uparrow \Rightarrow \text{move to unburnt region} \end{cases}$



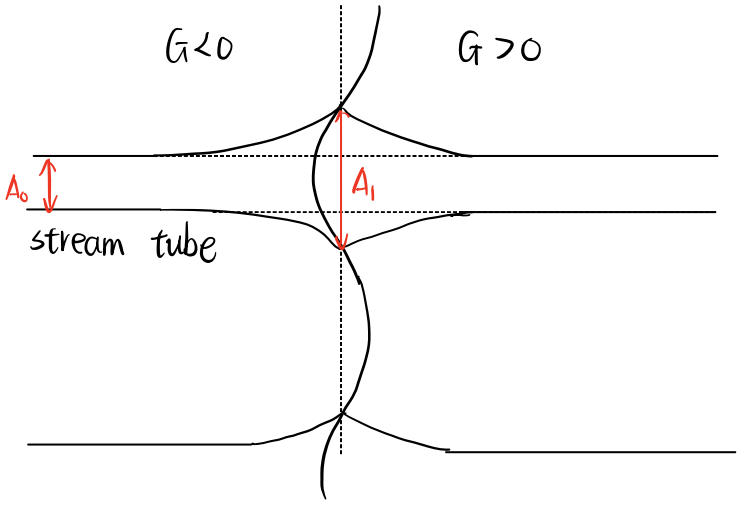
If $L < 0$, $\begin{cases} \kappa > 0 \Rightarrow S_L \uparrow \Rightarrow \text{move to unburnt region} \\ \kappa < 0 \Rightarrow S_L \downarrow \Rightarrow \text{move to burnt region} \end{cases}$



In conclusion, if $L < 0$, more curvature

if $L > 0$, tend to smooth

Flame instabilities : Hydrodynamics instability



heat release $\Rightarrow$ gas expansion $\Rightarrow$ deflection of a stream line

$\Rightarrow$ $\begin{cases} \text{stream tube cross aera} \uparrow \\ \text{stream tube cross aera} \downarrow \end{cases}$ $\Rightarrow$ $\begin{cases} u \downarrow \\ u \uparrow \end{cases}$

Analysis results in dispersion relation (How to get this formula ?)

$$\sigma = \frac{S_{L_0} k}{1+r}\left\{\sqrt{1+k^2L^2-\frac{2kL}{r}+\frac{1-r^2}{r^2}}-(1+kL)\right\}$$

where σ is the non-dimensional growth rate of the purturbation

$$\sigma = \frac{1}{f}\frac{df}{dt} = \frac{d\ln f}{dt}$$

r is density ratio

k is the wavenumber

$\Rightarrow$ perturbation grows exponentially in time only below a certain wavenumber $k^* = (r-1)/(2L)$

Comment:

① Without influence of curvature ($L = 0$), flame is unconditionally unstable.

② For perturbations at wave number $k > k^*$, a planar flame with positive Markstein number is unconditionally stable.

③ Burning velocity increases when flame front is concave and decrease when it is convex towards unburnt gas

⇒ initial perturbation become smoother.

Derivation of hydrodynamic instability